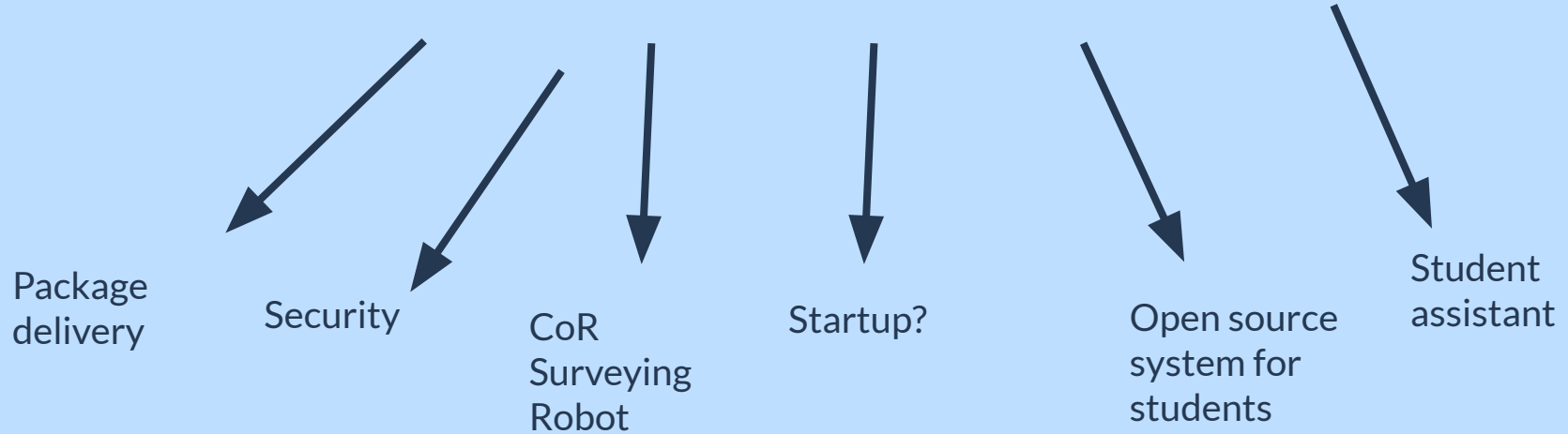


Project On-campus Vehicle Autonomous Launch (OVAL)



Overall Objective

- Develop a medium-scale autonomous vehicle to act as a test bed for autonomous vehicle research



Current Progress

NCDOT Research Symposium 2025

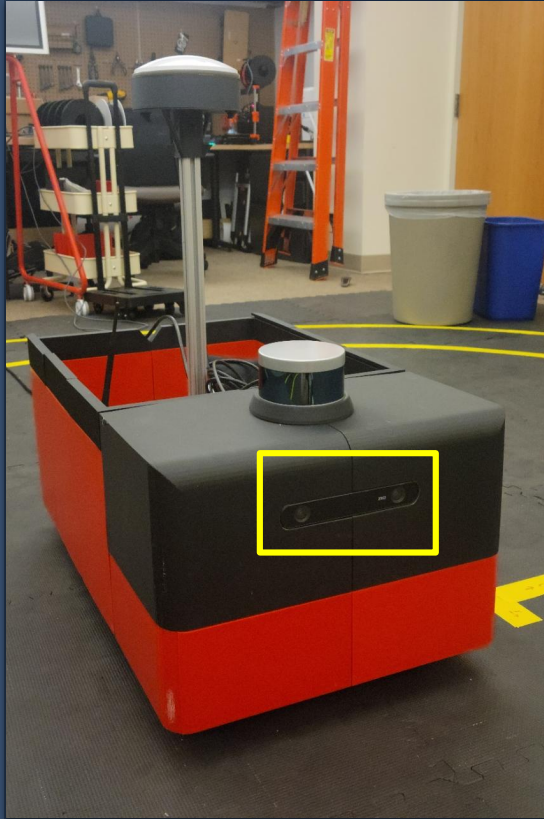


NC State COE Open House 2025



Sensor Capabilities

Stereo Camera



Spatial
Mapping

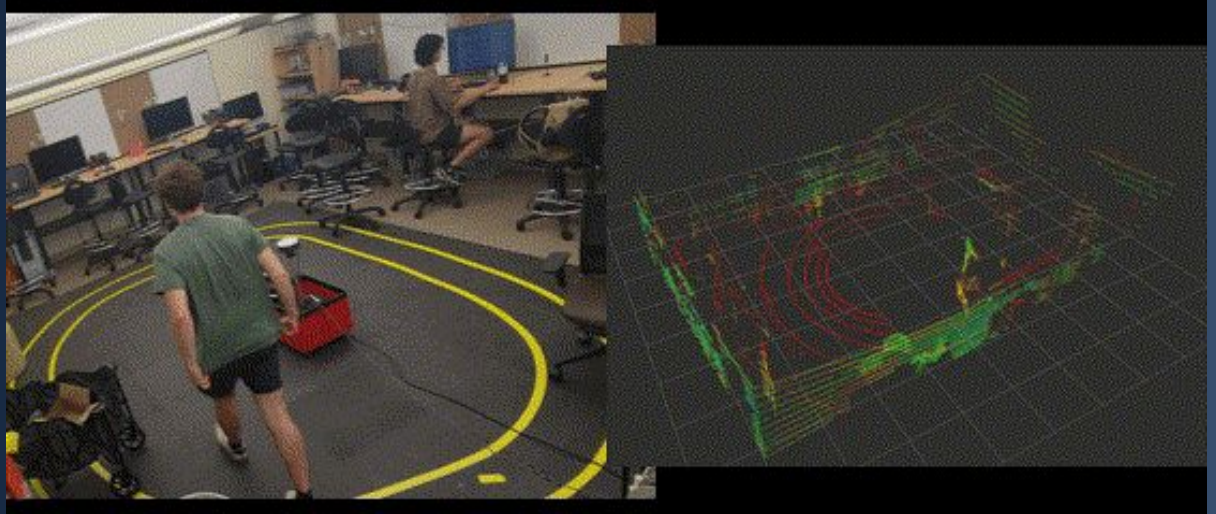
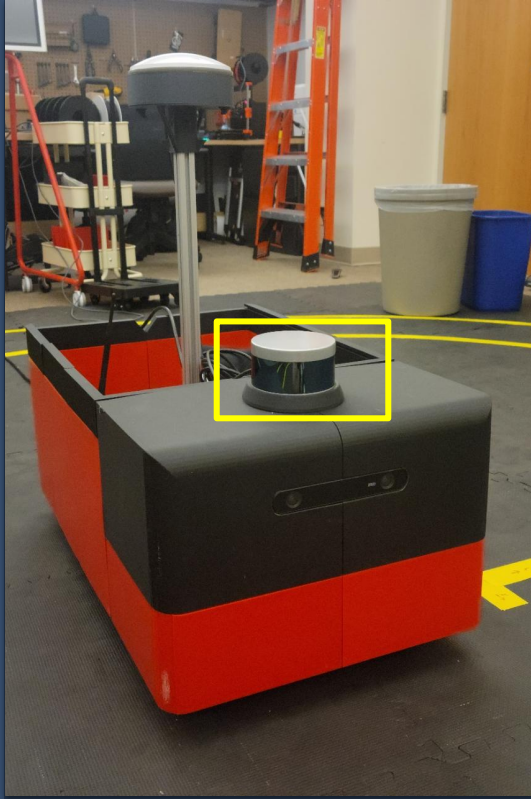


Positional
Tracking

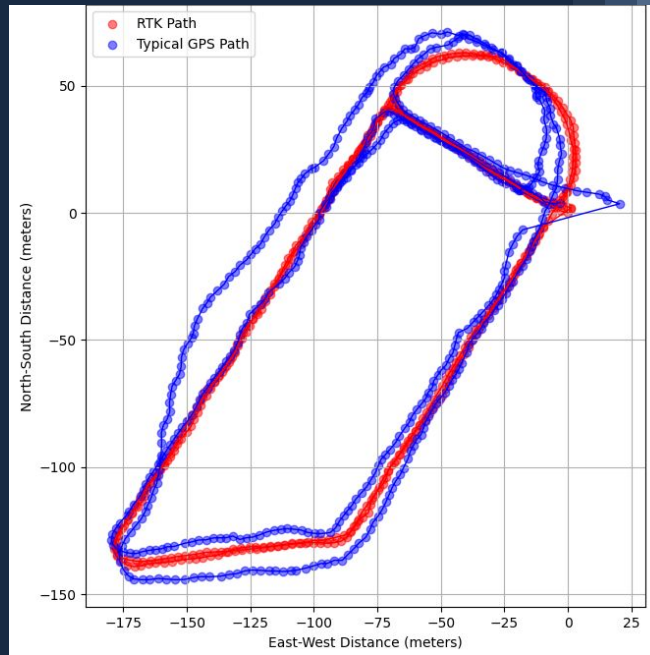


Body
Tracking

LiDAR



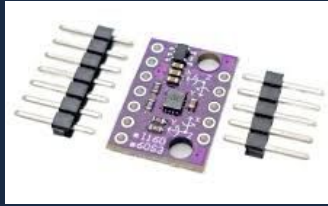
Real-Time Kinematic (RTK) GPS



Centimeter-level Accuracy

Other Sensors

6-axis IMU



Motor Encoders



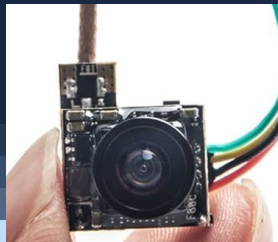
Radio Telemetry



Raspberry Pi Modem



Radio Camera



Cheap GPS



Web App



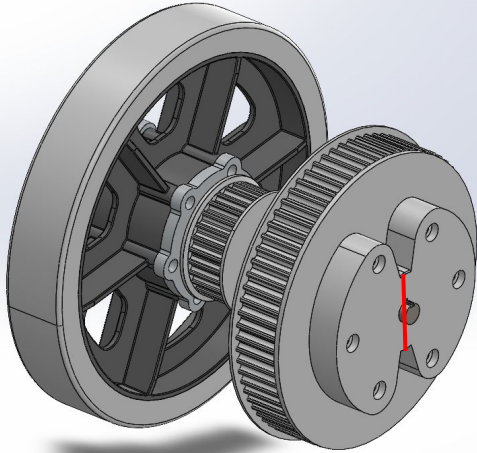
Semester Objective

SELF-DRIVING BY SEPTEMBER 30

1. Request the system to travel from the exterior of one building to another building on campus (EB 2 to Hunt)
2. The system is able to localize itself
3. Plans a route to take to achieve the objective
4. Travels along that route and stops when it detects an obstacle

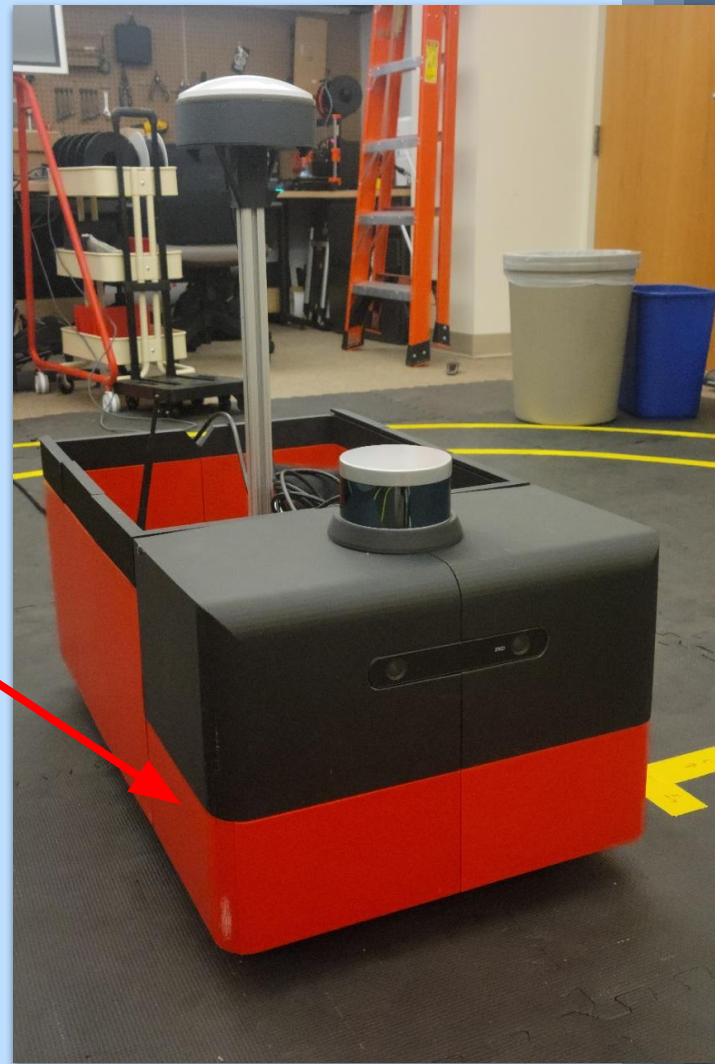
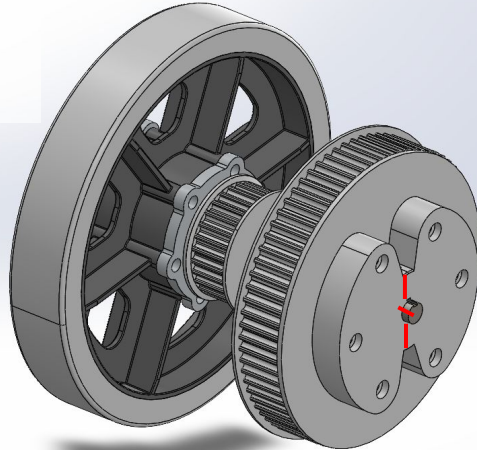


Mechanical

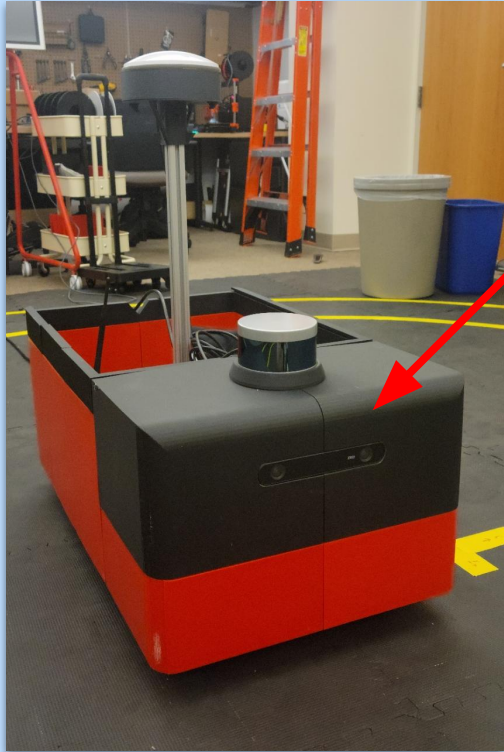


Shear stress snapped the **bolt** that connects the pulley to the axle

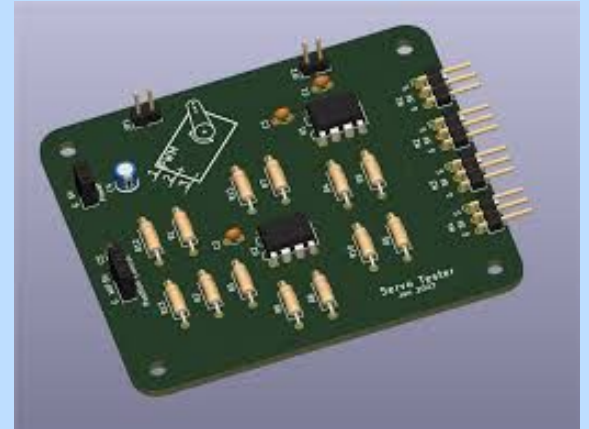
We have a solution, but it needs to be manufactured



Electrical

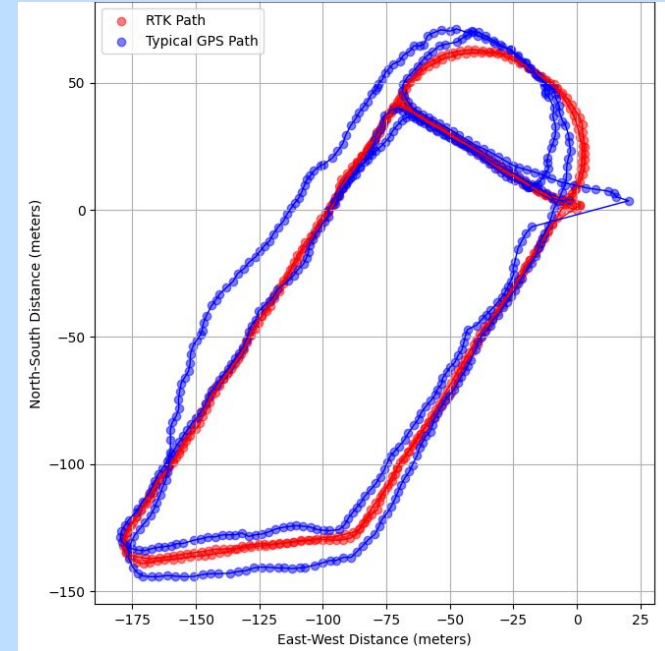


Wire Spaghetti



Software - Localization and Route Planning

- Use Google Maps API or OSM API to know possible paths
- Use RTK to know starting position
- Use route planning algorithm (Dijkstra, A*, Rapidly-exploring Random Tree)
- Provide a list of planned turns and waypoints for the path following algorithm



Software - Path Following

- Follow along a path from the route planning using computer vision with the ZED2i
- Properly handle detected intersections (going straight, turning, etc.)
- Stop when encountering an obstacle or pedestrian



Summary

- 3 Subteams
 - Mechanical
 - Electrical
 - Software
 - Localization and Route Planning
 - Path Following
- Weekly team meetings for updates and discussion (Mondays 6:30-7:00pm)
- Subteam meetings (specific to team)

